# Atlanta Regional Economic Modeling

Employment Growth and Regional Forecasting

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## 1 Purpose and Scope

This study examines employment growth and industry composition across metropolitan Atlanta.

The analysis will identify historical growth patterns, decompose industry employment changes, and develop a baseline regional employment forecast.

## 2 Data and Preparation

Primary sources: - BLS Quarterly Census of Employment and Wages - BEA Regional Economic Accounts - U.S. Census Bureau

## 2.1 BLS QCEW (Quarterly Census of Employment and Wages)

| Variable | Description |
|---|---|
| area_fips | Geographic identifier |
| own_code | Ownership classification |
| industry_code | NAICS industry classification |
| size_code | Establishment size classification |
| year | Reference year |
| disclosure_code | Confidentiality/suppression indicator |
| annual_avg_emplvl | Average monthly employment during the year |
| annual_avg_estabs | Average number of establishments |
| total_annual_wages | Total wages paid during the year |

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| | Unique | Missing Pct. | Mean | SD | Min | Median | Max | Histogram |
|---|---|---|---|---|---|---|---|---|
| area_fips | 1 | 0 | 13121.0 | 0.0 | 13121.0 | 13121.0 | 1.312100e+04 | |
| own_code | 5 | 0 | 4.6 | 1.1 | 0.0 | 5.0 | 5.000000e+00 | |
| agglvl_code | 9 | 0 | 77.0 | 1.2 | 70.0 | 77.0 | 7.800000e+01 | |
| size_code | 1 | 0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.000000e+00 | |
| year | 1 | 0 | 2024.0 | 0.0 | 2024.0 | 2024.0 | 2.024000e+03 | |
| annual_avg_estabs | 349 | 0 | 299.6 | 2791.6 | 0.0 | 18.0 | 7.271600e+04 | |
| annual_avg_emplvl | 765 | 0 | 3865.8 | 33586.0 | 0.0 | 187.5 | 9.481570e+05 | |
| total_annual_wages | 951 | 0 | 388967740.3 | 4471511847.3 | 0.0 | 14899954.0 | 9.558992e+10 | |
| taxable_annual_wages | 865 | 0 | 40680028.1 | 380231527.0 | 0.0 | 1143517.5 | 9.987105e+09 | |
| annual_contributions | 864 | 0 | 476425.1 | 4472996.6 | 0.0 | 11477.0 | 1.166167e+08 | |
| annual_avg_wkly_wage | 809 | 0 | 1245.6 | 1125.0 | 0.0 | 1222.0 | 1.003500e+04 | |
| avg_annual_pay | 947 | 0 | 64771.8 | 58501.9 | 0.0 | 63554.0 | 5.218110e+05 | |
| lq_annual_avg_estabs | 268 | 0 | 1.2 | 4.0 | 0.0 | 0.8 | 8.290000e+01 | |
| lq_annual_avg_emplvl | 303 | 0 | 1.2 | 5.4 | 0.0 | 0.4 | 1.090000e+02 | |
| lq_total_annual_wages | 276 | 0 | 1.0 | 4.4 | 0.0 | 0.3 | 8.420000e+01 | |
| lq_taxable_annual_wages | 278 | 0 | 0.8 | 3.2 | 0.0 | 0.2 | 1.131000e+02 | |
| lq_annual_contributions | 262 | 0 | 0.8 | 4.0 | 0.0 | 0.2 | 1.090000e+02 | |
| lq_annual_avg_wkly_wage | 132 | 0 | 0.6 | 0.4 | 0.0 | 0.7 | 2.500000e+00 | |
| lq_avg_annual_pay | 132 | 0 | 0.6 | 0.4 | 0.0 | 0.7 | 2.500000e+00 | |
| oty_annual_avg_estabs_chg | 185 | 0 | 36.4 | 413.3 | -3281.0 | 2.0 | 8.865000e+03 | |
| oty_annual_avg_estabs_pct_chg | 433 | 0 | 25.9 | 39.9 | -100.0 | 14.6 | 4.000000e+02 | |
| oty_annual_avg_emplvl_chg | 434 | 0 | 43.5 | 629.4 | -6040.0 | 0.0 | 1.107900e+04 | |
| oty_annual_avg_emplvl_pct_chg | 415 | 0 | 4.7 | 25.8 | -100.0 | 0.0 | 4.333000e+02 | |
| oty_total_annual_wages_chg | 910 | 0 | 23016378.9 | 199829528.4 | -336362149.0 | 76437.0 | 5.566331e+09 | |
| oty_total_annual_wages_pct_chg | 476 | 0 | 7.6 | 25.7 | -100.0 | 0.9 | 3.891000e+02 | |
| oty_taxable_annual_wages_chg | 835 | 0 | -1103894.9 | 14334959.8 | -270528317.0 | 0.0 | 3.158488e+07 | |
| oty_taxable_annual_wages_pct_chg | 456 | 0 | 3.2 | 30.1 | -100.0 | 0.0 | 6.767000e+02 | |
| oty_annual_contributions_chg | 828 | 0 | 14289.7 | 235053.7 | -1382249.0 | 0.0 | 4.070233e+06 | |
| oty_annual_contributions_pct_chg | 598 | 0 | 8.0 | 63.7 | -95.6 | 0.0 | 1.106000e+03 | |
| oty_annual_avg_wkly_wage_chg | 430 | 0 | 48.2 | 190.6 | -2450.0 | 6.0 | 1.848000e+03 | |

| | Unique | Missing Pct. | Mean | SD | Min | Median | Max | Histogram |
|---|---|---|---|---|---|---|---|---|
| oty_annual_avg_wkly_wage_pct_chg | 317 | 0 | 2.9 | 9.5 | -100.0 | 0.4 | 8.730000e+01 | |
| oty_avg_annual_pay_chg | 895 | 0 | 2507.2 | 9910.3 | -127388.0 | 321.0 | 9.609800e+04 | |
| oty_avg_annual_pay_pct_chg | 319 | 0 | 2.9 | 9.5 | -100.0 | 0.4 | 8.720000e+01 | |

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Table 1: Descriptive Stats for BLS QCEW

| | N | Mean | SD | Min | Median | Max |
|---|---|---|---|---|---|---|
| area_fips | 2184.0 | 13121.0 | 0.0 | 13121.0 | 13121.0 | 1.312100e+04 |
| own_code | 2184.0 | 4.6 | 1.1 | 0.0 | 5.0 | 5.000000e+00 |
| agglvl_code | 2184.0 | 77.0 | 1.2 | 70.0 | 77.0 | 7.800000e+01 |
| size_code | 2184.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.000000e+00 |
| year | 2184.0 | 2024.0 | 0.0 | 2024.0 | 2024.0 | 2.024000e+03 |
| annual_avg_estabs | 2184.0 | 299.6 | 2791.6 | 0.0 | 18.0 | 7.271600e+04 |
| annual_avg_emplvl | 2184.0 | 3865.8 | 33586.0 | 0.0 | 187.5 | 9.481570e+05 |
| total_annual_wages | 2184.0 | 388967740.4 | 3471511847.3 | 0.0 | 14899954.0 | 9.558992e+10 |
| taxable_annual_wages | 2184.0 | 40680028.1 | 380231527.0 | 0.0 | 1143517.5 | 9.987105e+09 |
| annual_contributions | 2184.0 | 476425.1 | 4472996.6 | 0.0 | 11477.0 | 1.166167e+08 |
| annual_avg_wkly_wage | 2184.0 | 1245.6 | 1125.0 | 0.0 | 1222.0 | 1.003500e+04 |
| avg_annual_pay | 2184.0 | 64771.8 | 58501.9 | 0.0 | 63554.0 | 5.218110e+05 |
| lq_annual_avg_estabs | 2184.0 | 1.2 | 4.0 | 0.0 | 0.8 | 8.290000e+01 |
| lq_annual_avg_emplvl | 2184.0 | 1.2 | 5.4 | 0.0 | 0.4 | 1.090000e+02 |
| lq_total_annual_wages | 2184.0 | 1.0 | 4.4 | 0.0 | 0.3 | 8.420000e+01 |
| lq_taxable_annual_wages | 2184.0 | 0.8 | 3.2 | 0.0 | 0.2 | 1.131000e+02 |
| lq_annual_contributions | 2184.0 | 0.8 | 4.0 | 0.0 | 0.2 | 1.090000e+02 |
| lq_annual_avg_wkly_wage | 2184.0 | 0.6 | 0.4 | 0.0 | 0.7 | 2.500000e+00 |
| lq_avg_annual_pay | 2184.0 | 0.6 | 0.4 | 0.0 | 0.7 | 2.500000e+00 |
| oty_annual_avg_estabs_chg | 2184.0 | 36.4 | 413.3 | -3281.0 | 2.0 | 8.865000e+03 |
| oty_annual_avg_estabs_pct_chg | 2184.0 | 25.9 | 39.9 | -100.0 | 14.6 | 4.000000e+02 |
| oty_annual_avg_emplvl_chg | 2184.0 | 43.5 | 629.4 | -6040.0 | 0.0 | 1.107900e+04 |
| oty_annual_avg_emplvl_pct_chg | 2184.0 | 4.7 | 25.8 | -100.0 | 0.0 | 4.333000e+02 |
| oty_total_annual_wages_chg | 2184.0 | 23016378.9 | 199829528.4 | -336362149.0 | 76437.0 | 5.566331e+09 |
| oty_total_annual_wages_pct_chg | 2184.0 | 7.6 | 25.7 | -100.0 | 0.9 | 3.891000e+02 |
| oty_taxable_annual_wages_chg | 2184.0 | -1103894.9 | 14334959.8 | -270528317.0 | 0.0 | 3.158488e+07 |
| oty_taxable_annual_wages_pct_chg | 2184.0 | 3.2 | 30.1 | -100.0 | 0.0 | 6.767000e+02 |
| oty_annual_contributions_chg | 2184.0 | 14289.7 | 235053.7 | -1382249.0 | 0.0 | 4.070233e+06 |
| oty_annual_contributions_pct_chg | 2184.0 | 8.0 | 63.7 | -95.6 | 0.0 | 1.106000e+03 |
| oty_annual_avg_wkly_wage_chg | 2184.0 | 48.2 | 190.6 | -2450.0 | 6.0 | 1.848000e+03 |
| oty_annual_avg_wkly_wage_pct_chg | 2184.0 | 2.9 | 9.5 | -100.0 | 0.4 | 8.730000e+01 |
| oty_avg_annual_pay_chg | 2184.0 | 2507.2 | 9910.3 | -127388.0 | 321.0 | 9.609800e+04 |
| oty_avg_annual_pay_pct_chg | 2184.0 | 2.9 | 9.5 | -100.0 | 0.4 | 8.720000e+01 |

#### 2.1.1 BLS QCEW:

Dataset: Quarterly Census of Employment and Wages (QCEW) Source: US BLS Study period: 2022 to 2025 using annual data Initial geography: Fulton County, Georgia (FIPS 13121), expandable to 21 counties

within ARC's region, and US as a national benchmark.

QCEW primarily provides administrative employment and wage statistics for jobs covered by unemployment insurance and federal civilian employment programs. We will thus focus on: - Employment growth -Industry composition -Wage trends -Atlanta's economic performance relative to national trends (needs to be defined)

| Variable | Full Name | Purpose |
|---|---|---|
| `area_fips` | Geographic FIPS Identifier | Identify counties |
| `year` | Reference Year | Track annual changes |
| `industry_code` | NAICS Industry Code | Compare industry composition |
| `annual_avg_emplvl` | Annual Average Employment | Measure employment growth |
| `annual_avg_estabs` | Annual Average Establishments | Track establishment counts |
| `total_annual_wages` | Total Annual Wages | Analyze compensation trends |

#### 2.1.2 Summarizing observations

The intiial review of the QCEW dataset indicates that:

- **Employment:**
- **Data issues:**
- **Industry classifications:**

### 2.2 Prepare County-Level Data

Keep industry observations separate from all-industry totals to avoid double counting.

## 3 Aggregate Regional Employment

Construct regional industry totals by summing reported county employment. Track coverage to identify suppressed or missing county-industry observations.

## 4 5. Save Prepared Datasets

## 5 6. Initial Data Review

## 6 7. Data Notes

- QCEW records covered jobs, not unique employed residents.
- All-industry totals are retained separately from NAICS sectors.
- Some county-industry values may be suppressed.
- Aggregated industry employment may therefore be incomplete.
- The 2022–2025 dataset provides an initial comparable industry series; a longer history is needed for forecasting.
- Full-county aggregation does not represent the exact boundaries of every ARC transportation planning geography.

Data source: https://www.bls.gov/cew/additional-resources/open-data/